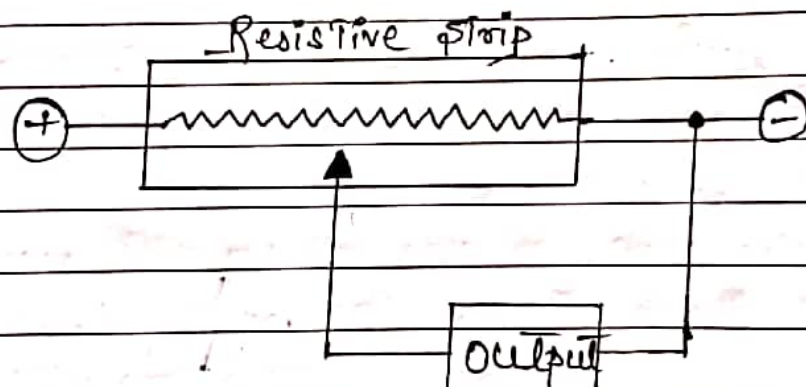


Potentiometer - It is a three Terminal device with rotating or sliding contact which forms adjustable voltage divider.

It functions as variable resistor or Rheostat when two terminals are used (one end and wiper). Potentiometer has wide variety of applications such as volume or tone control in audio system, as position sensor (or transducer) in joystick etc.

Linear Potentiometer



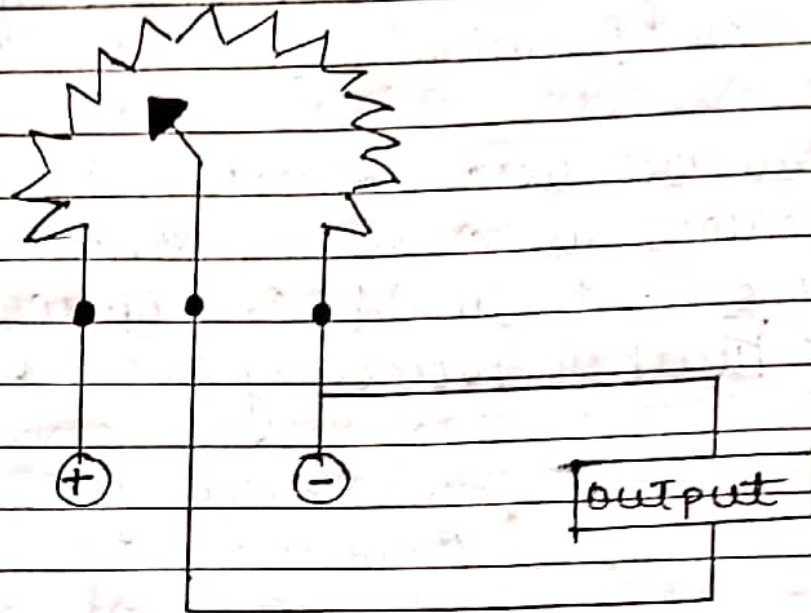
- Linear Potentiometer contains electrical resistance in the form of straight section of track. The track may consist of strip of resistive Polymer or insulator with coil of Nichrome wire wrapped around it.
- For sensing application, Potentiometer is used as voltage divider. For the same, fixed potential is used applied across its full length as shown in fig.

- As shown Wiper slides along the track, hence sensing Voltage varies linearly with Position of the wiper. Hence it is known as linear Potentiometer.
- It can be used directly to control analog indicator in a analog meter. It can also be processed by ADC (Analog To Digital Converter) for Further Processing.
- Ex - Linear Potentiometer from Companies such as Bourns.

Advantages - Simple, inexpensive, compact requires few components.

Disadvantages - Though the track contains lubricant, some ^{where} ~~where~~ such occurs as a result of motion of the Wiper contact.

Rotary Potentiometer :-



- Rotary Potentiometer is categorized into arc-segment rotary Potentiometer and multiturn rotary Potentiometer.
- Rotary Potentiometer of arc segment type. This type is most common compare to multiturn type.
- As shown, it contains Resistor in the shape of arc, referred as track. It may be strip of resistive Polymer or may consists of Insulator with Coil of Nichrome wire wrapped around it.
- Wiper slides along the track, sensing a voltage that varies linearly with angular position of the wiper.

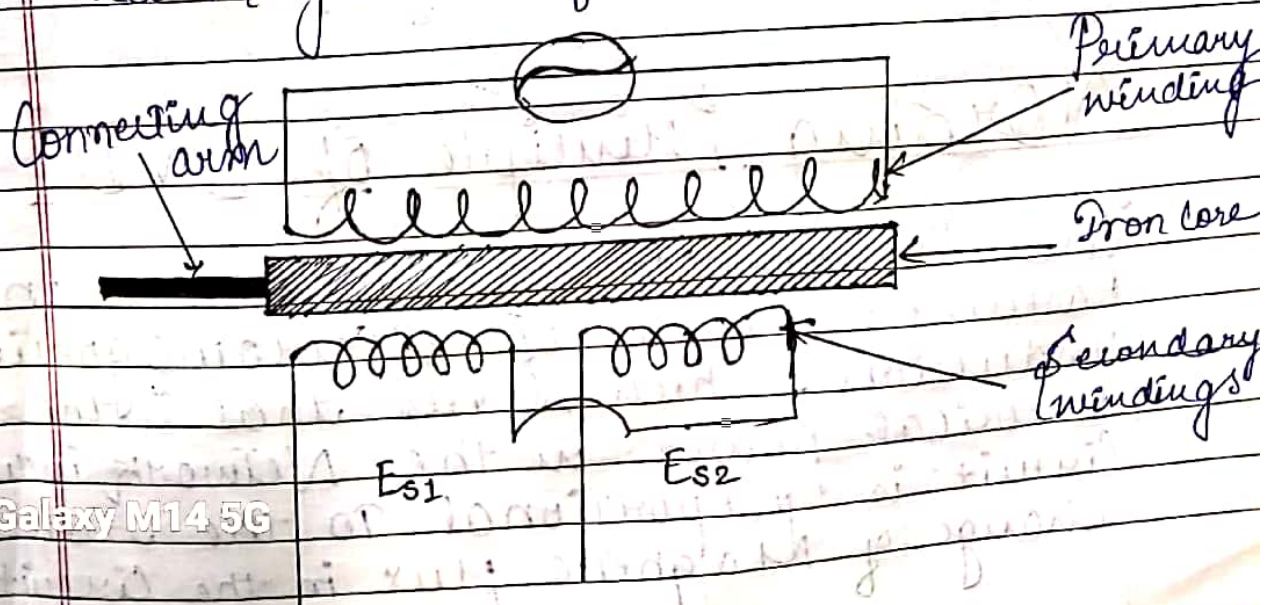
Advantages - Simple, Inexpensive, Compact, requires few components.

Disadvantages - Though the track contains lubricant, some wear occurs as a result of motion of the wiper contact.

LVDT TRANSDUCER

LVDT stands for Linear Variable Differential Transformer. It converts the linear motion into an electrical signal using an inductive transducer.

It is one of the major types of Inductive Transducer. LVDTs are considered the most accurate inductive sensors that measure displacement according to the polarity and magnitude of the Net induced EMF.



Construction of LVDT :-

The transformer and LVDT share a similar construction. It consists of one Primary Windings (P) and two Secondary windings (S_1 & S_2). The primary and secondary windings are bounded by a hollow cylinder, known as the former.

The primary windings is at the center and the Secondary windings are present on both sides of the Primary winding at an equal distance from the center. Both the

Both the Secondary windings have an equal no. of turns and they are linked with each other in series opposition, i.e. they are wound in opposite directions, but are connected in series with each other.

Working Principle of LVDT

The Working of LVDT is based on Faraday's Law of Electromagnetic Induction, which states that "the electrical power in the Network induced circuit is proportional to the rate of change of Magnetic flux in the circuit".

As the primary windings of LVDT is connected to the AC power supply, the alternating Magnetic field is produced in the Primary windings, which results in the induced EMF of Secondary Windings.

Let's assume that the induced voltages in the Secondary windings S_1 & S_2 are E_1 & E_2 respectively. Now according to Faraday's Law, the rate of change of Magnetic flux i.e. $\frac{d\phi}{dt}$ is directly proportional to the Magnitude of induced EMF i.e. E_1 and E_2 .

The output voltage E_0 in the circuit is given by $E_0 \rightarrow E_1 - E_2$.

Characteristics of LVDT - / Specifications -

- Linearity
- Sensitivity
- Resolution
- Electrical output
- Temperature Range etc.

Advantages of LVDT -

1. High output
2. low hysteresis.
3. low electrical Noise
4. less power consumption

Disadvantages of LVDT -

1. Since LVDT is an inductive Transducer, it is sensitive to the stray magnetic field, hence an extra setup is required to protect from stray Magnetic field.
2. As LVDT is an electromagnetic device, it is also affected by vibrations and Temperature.

Applications of LVDT -

1. It is used in industries in the field of Automation, aircraft, Turbines, Satellite, Hydraulics etc.
2. LVDT is used to measure Physical Quantity such as force, Tension, Pressure, weight etc. here LVDT is used as secondary Transducer.

3. LVDT plays important role in Geotunnel Instrumentation.
4. LVDT plays an important role in the Marine and offshore.
5. It also plays imp. role in Power Generation.

EDDY CURRENT TYPE TRANSDUCER

This type of Transducer is comparatively low in the measurement field and depends mostly on the quality of a high alternating source which is fed to a set of coils. one coil is called the active coil and the other provides temperature compensation (compensating coil) by being the adjacent arm of a bridge circuit. A conducting material is kept close to the active coil so as to make it influenced by its absence or presence, or, by being any closer or away. Magnetic flux is induced in the active coil and is passed through the conductor producing eddy currents. The density of this current will be maximum at the surface and will lessen as the depth increases. This penetration depth can be calculated by using the Equation -

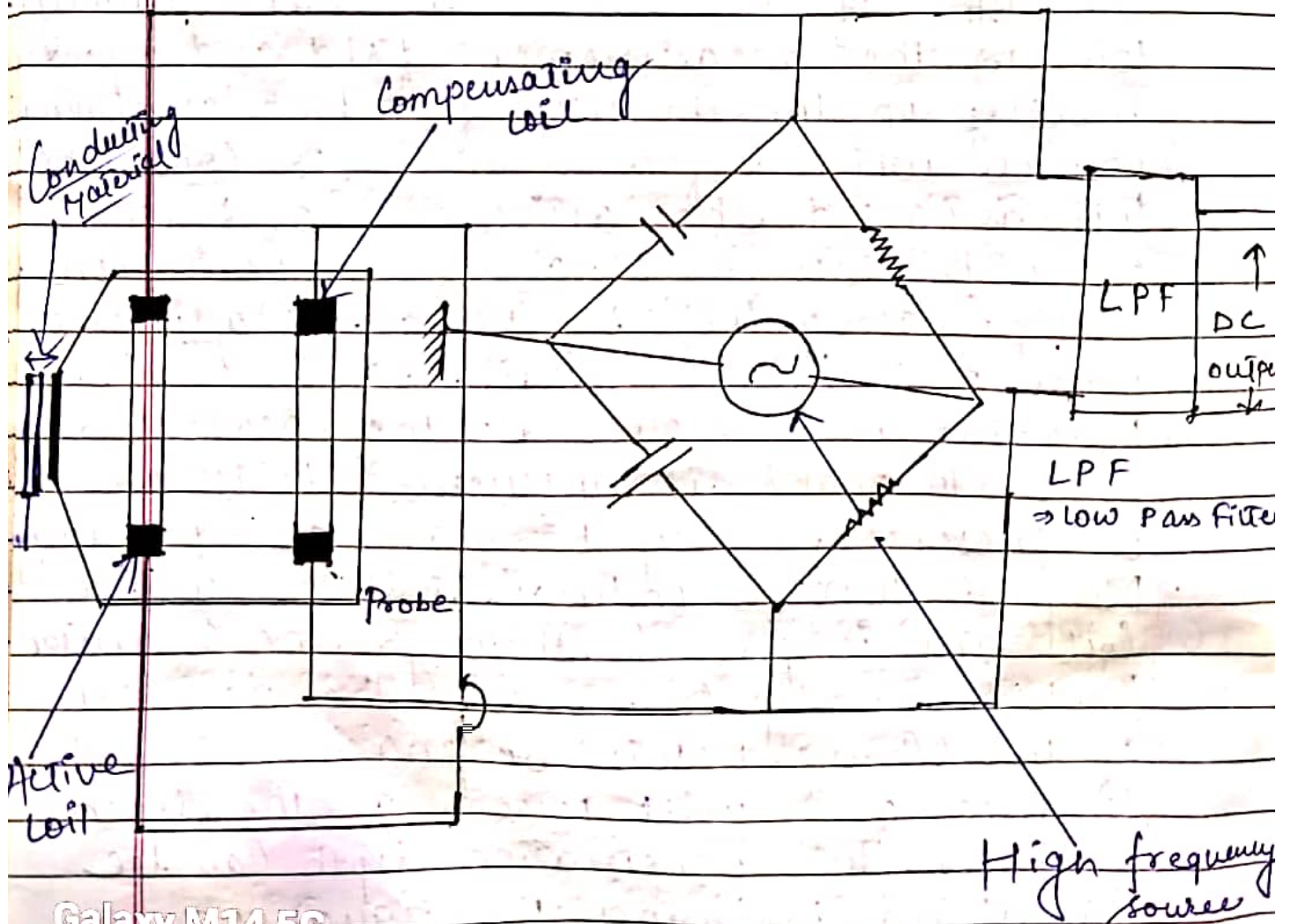
$$\delta \Rightarrow \frac{1}{f \pi \mu \sigma}$$

$\delta \Rightarrow$ Penetration Depth (m)

$f \Rightarrow$ frequency (Hz)

$\mu \Rightarrow$ Magnetic Permeability

$\sigma \Rightarrow$ Electrical Conductivity (S/m)



Working - The active coil is kept closer to the conducting material and both of them are placed inside a probe. The compensating coil is kept further away from the conducting material. The high frequency source acts as the bridge circuit and feeds the coil across the two capacitors. The amount of eddy current produced becomes more as the distance between the conducting material and the active coil become less. This causes a change in the impedance of the active coil and thus imbalance the bridge circuit.

The bridge circuit produces an output proportional to the amount of closeness between the conducting material and the active coil. The output of the bridge circuit is given to a low pass filter (LPF) and then dc output is calculated. The high frequency allows a thin target to be used and also with this, the frequency responses become good up to a target frequency 1/10th the supply frequency.

Application - Since it is a non-contact device, it is suitable for higher resolution measurement applications. The device is used for finding out the position of an object that is conductive in nature.

- Position Measurement.
- Vibrating motion measurement.

Advantages -

1. Measurement of distance can be carried out even in rough or mixed environment.
2. Cost-effective.
3. The device is insensitive to material in the gap between the probe and the conductor.
4. The device is less expensive.

Disadvantages -

1. The result will be precise only if the gap between the probe and the conductor is small.
2. The device cannot be used for finding the position of non-conductive materials.

3. The device is highly Temperature sensitive.
4. There always occurs a non-linear relationship between the distance and impedance of the active coil.

Capacitance Pickups -

Application -

The Capacitive Pick-up Probe is used for observation of the Particle beam micro-structure of R-F acceleration. The Probe should be followed by a broadband signal processing system.

- Time of flight Measurement for determination of Particle Energy.
- Longitudinal Emittance measurements.
- Determination of Minimum acceleration voltage for R-F Structure (Phase-Scan).
- Measurement and Monitoring of stability of an accelerator with respect to the Energy.

Principle -

By passing the inner electrode, the Particle bunch induces a current on the electrode which is proportional to the time derivation of intensity distribution within the bunch.

Piezoelectric Ultrasonic Transducer

Piezoelectric Transducer are a type of electroacoustic Transducer that converts the electrical charges produced by some forms of solid material into Energy. The word "Piezoelectric" literally means electricity caused by Pressure.

Piezoelectric ultrasonic Transducer generate ultrasonic activity, meaning it measures the Pressure of sound waves above the frequencies that can be heard by the human ear. It functions by rapidly expanding and contracting when appropriate electrical frequency and voltage is applied.

Typically used in cleaning system, the expansion and contraction causes the ultrasonic cleaner Transducer's diaphragm, which serves as the Pressure-sensing element of the Unit, to vibrate introducing ultrasonic activity into the cleaning tank in the process. The Piezoelectric ultrasound Transducer also offers the advantages of high ~~electro~~ electroacoustic efficiency while minimizing heat generation.

Advantages of Piezoelectric Transducers -

- Self-Generating - Due to their material ability to produce voltage while under the influence of some energy, Piezo Transducer circuits do not require an external power source.
- Easily Accessible - Due to their small dimensions and large measuring range, are easy to handle, install and use.
- High-Frequency Response - A much higher than-normal frequency response means the Parameters of these Transducers shifts Quickly.
- Flexibility - Due to how most of the materials used in construction can be shaped into different shapes and sizes, you can apply these Transducers in various fields.

Disadvantages of Piezoelectric Transducers

- Small Electric Charge - While they might be self-generating, you'll require a high impedance cable to establish a connection with an electrical interface.

Affected by Various environment and conditions - Piezo Transducers cannot measure output in the static condition, and temperature and humidity variations can affect output.

- Low Output - While some materials produce output rates that are relatively higher than their counterparts, output is still low.

Hall Effect Transducer

- The Hall effect element is a Transducer used for measuring the magnetic field by converting it into an EMF.
- The direct measurement of the Magnetic field is not possible. Thus, the Hall effect Transducer is used for indirect measurement of Magnetic field.
- The Transducer converts the Magnetic field into an electric Quantity which is easily measured by the analog and Digital meters is called hall effect Transducer.

Working Principle :-

- The Working Principle of Hall effect Transducer is based on Hall effect. It was discovered by Edwin Hall in 1879.
 - Hall Effect is basically the process of development of Potential difference across the two faces of a current carrying strip when the strip is kept in a Magnetic field.
 - The magnitude of voltage depends upon the current, strength of magnetic field and the property of conducting material.
- Galaxy India Hall effect is found in conducting material

and Semiconductor in varying amount depending upon the density and mobility of current carrier.

- The hall hall effect elements are based on thin film of semiconducting material like indium Arsenide.

$$E_H = K_H \frac{I B}{t}$$

K_H = Hall coefficient

I = current

B = flux density

t = thickness

Advantages

Application of Hall Effect Transducer-

1. It has long life because no wear.
2. They are highly reliable.
3. It has very high-speed operation.
4. They can be operated in a wide Temp. range.

Disadvantages of H.E.T-

1. It cannot measure distance more than 10cm.
2. Strong magnetic field can cause inaccurate reading.

Application of H. E. T -

1. The Hall effect element is used for converting the magnetic flux into an electric Transducer.
2. The Hall effect element measures the displacement of the structural element.
3. The Hall effect Transducer is also used for measuring the current without any physical connection between the conductor circuit and meter.
4. The Hall effect Transducer is used for measuring the Power of the Conductor.
5. open / close of laptop flip Screen.

Optical Transducer

Page No:
Date:

The optical transducer convert light into electrical quantity. They are also called as photoelectric transducer. The optical transducer can be classified as photo emission, photoconductive and photovoltaic transducer. The photo emission devices operate on the principle that radiation falling on a cathode causes electrons to be emitted from the cathode surface.

An optical transducer converts light rays into an electronic signal. The purpose of an optical transducer is to measure a physical quantity of light and depending on the type of transducer, then translates it into a form that is readable by an integrated measuring devices.